

Seat No.: _____

Enrollment No.: _____

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (CL/EC/ME) Sem - I Examination, 2009-10

CL - 101 Fundamentals of Civil Engineering

Date: 15.12.2009, Tuesday

Time: 02:p.m. To 05:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Differentiate plane surveying and geodetic surveying. [4]

(b) Define the following terms: [3]

(i) Magnetic Meridian (ii) Back Sight (iii) Fore Bearing & Back Bearing

Q - 2 (a) The following bearings were observed while traversing with compass: [6]

Line	F. B.	B. B.
AB	124° 30'	304° 30'
BC	68° 15'	249° 45'
CD	312° 00'	127° 15'
DA	197° 00'	20° 15'

Find which stations are affected by local attraction? Determine the corrected bearings.

(b) Write short notes on: [4]

(i) Methods of ranging (ii) Errors in chaining

(c) (i) Which are the uses of planimeter? [4]

(ii) The area of the figure was measured with planimeter, where the tracing arm being set to the natural scale. The initial and final readings were observed to be 2.289 and 7.628 respectively. The anchor point was fixed outside the figure; the zero on the disc passed the index mark two times in clockwise direction and once in anti-clockwise direction and the multiplying factor is 100 cm².

OR

Q - 2 (a) The following staff readings were taken with dumpy level and 4 m leveling staff: [6]

0.365, 0.405, 0.635, 1.765, 1.905, 2.360, 1.790, 0.355, 0.695, 1.240 and 2.160.

The instrument was shifted after 4th and 7th reading. The R.L. of first reading is 100.000 m. Calculate the R.L. of all the points by suitable method. Also apply arithmetic checks.

- (b) Following are the bearings of lines of a closed traverse ABCD. [5]

Line	F. B.
AB	N 50° E
BC	N 80° E
CD	S 40° W
DA	N 70° W

Find the included angles and apply the necessary checks.

- (c) Explain the general principles of surveying in detail. [3]

Q - 3 (a) List out various aids and devices used to control, regulate and guide the traffic. Explain traffic signals in detail. [4]

- (b) Write a short note on 'BOT Projects'. [3]

- (c) (i) Discuss the correction for temperature and correction for sag. [7]

(ii) The distance between two stations was 1500 m when measured with a 20 m chain. The same distance when measured with a 30 m chain was found to be 1495 m. If the 20 m chain was 0.06 m too long, what was the error in the 30 m chain?

OR

Q - 3 (a) What is role of transportation in national development? Which are the different modes of the transportation? [4]

- (b) What are the merits and demerits of Railways? [3]

- (c) (i) Explain the correction for absolute length and correction for pull. [7]

(ii) A steel tape was standardised as 30 m at 18°C temperature. A line was measured as 540.00 m with temperature during measurement as 30°C. Calculate true length of chain, if coefficient of expansion is 0.000012 / °C rise of temperature.

SECTION - II

Q - 4 (a) Draw a neat sketch of a cross section of wall showing various building components. [5]

- (b) Discuss the scope of Civil Engineering. [2]

Q - 5 (a) Draw an electrical installation **plan** for a bed room of size of 5 m × 4 m. Also prepare the schedule giving relevant descriptions of doors, windows and [6]

ventilators.

- (b) Draw a layout for an industrial building of your choice. [4]
- (c) Enlist various branches of civil engineering and explain briefly the importance of each branch. [4]

OR

- Q - 5** (a) Draw the detailed **plan** of a room having a size of 4 m × 3 m. Also show the details of doors and windows. [5]
- (b) Classify the buildings based upon occupancy. [5]
 - (c) Enlist various principles of building planning and explain roominess and circulation in detail. [4]

- Q - 6** (a) What is canal? Classify the canals in different ways. [4]
- (b) Write short notes on: [6]
 - (i) Water requirements for different uses
 - (ii) Conservation of water
 - (c) What are the basic requirements of building planning? [4]

OR

- Q - 6** (a) Discuss various types of sources of water. [4]
- (b) Explain Hydrologic Cycle with neat sketch. Which are the applications of hydrology? [6]
 - (c) Write short notes on: [4]
 - (i) Different types of loads acting on structure
 - (ii) Prospect and Elegance
